

ChangeNet AI: Intelligent Network Monitoring and Management

Network Management System

Submitted by

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ABSTRACT

ChangeNet AI is a lightweight web-based network monitoring and device analysis system developed to provide a simple and user-friendly environment for demonstrating network monitoring concepts. The system is primarily designed for academic and small-scale demonstration purposes, where users need to maintain information about network devices and understand their current status and monitoring behaviour without the complexity of enterprise-level network management platforms. The application provides an administrative login interface followed by a centralized dashboard containing major sections such as device management, network monitoring, alerts, smart analysis, device comparison, and monitoring reports. Users can maintain a collection of manually registered network devices containing information such as device name, IP address, device type, and current status. The system also provides predefined demonstration devices to make the application easier to evaluate and present. The Network Monitoring module displays simplified monitoring parameters such as CPU utilization, memory utilization, latency, packet loss, incoming traffic, and outgoing traffic. These values are generated as simulated monitoring data for demonstration purposes rather than being collected from real network devices. A graphical representation of incoming and outgoing traffic is provided along with recent monitoring measurements. The Alert Center provides sample network events such as router response delay, switch port warnings, and device disconnection. These alerts demonstrate how network events can be visually classified according to their priority. The project also includes a Smart Analysis section containing a Change Risk Indicator, Device Health Timeline, and Smart Change Insight. The Device Stability Score provides a simple rule-based numerical indication of device stability. Based on the available device status information, the system assigns a score within a 0–100 range and classifies the corresponding change risk as Low, Medium, or High. Smart Change Insight converts this information into an understandable textual interpretation. The Compare Devices module allows two devices to be compared using their current status, stability score, and risk level. The Monitoring Report module generates a presentation-ready summary containing important device and stability information. The frontend of the application is developed using HTML, CSS, and JavaScript, while Node.js is included as a lightweight local backend environment. The application does not require a database or complex external framework. The current implementation focuses on simplicity, usability, visual clarity, and academic demonstration while showing how basic monitoring information can be transformed into useful analytical information.

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COURSE OUTCOMES

On completion of course, students will be able to

- Explain mobile network management using TMN, SNMP, FCAPS, and model-driven architectures.
- Analyze and implement alarm and fault management using SDN, REST APIs, and virtualization frameworks.
- Perform CRUD and telemetry operations using YANG, NETCONF, RESTCONF, and related tools like NETCONF4J.
- Fault, Performance mgmt. life-cycle, application use cases
- Over view of AI-Driven Network Service Orchestration in OSS layer of TMN

CHAPTER 1

INTRODUCTION

1.1 MOTIVATION

Network-connected devices play an important role in modern communication and information systems. Devices such as routers, switches, servers, access points, and computers must remain available for users and applications to operate effectively. Monitoring the condition of these devices helps administrators understand the current state of a network and identify situations that may require attention. Professional network monitoring platforms provide extensive monitoring capabilities, but their configuration and functionality can be complex for beginners. For an academic project, there is a need for a simpler environment through which students can understand important concepts such as device management, status monitoring, alerts, performance information, and basic device analysis.

ChangeNet AI was developed to address this requirement. The project provides a centralized web interface through which users can manage network devices and view simplified monitoring information. In addition to basic monitoring concepts, the system introduces simple analytical features such as Device Stability Score, Change Risk Indicator, Smart Change Insight, Device Health Timeline, Device Comparison, and Monitoring Report generation. The project therefore combines fundamental networking concepts with web application development in a form that is easy to understand, demonstrate, and evaluate.

1.2 EXISTING SYSTEM

Existing network monitoring platforms are generally designed to monitor large or complex network environments. Such systems may provide features including device discovery, availability monitoring, performance monitoring, historical information, alert management, dashboards, reporting, and network visualization. Platforms such as Zabbix, Nagios, and other professional monitoring solutions demonstrate how network resources can be continuously monitored and analyzed. However, these platforms may involve configuration requirements and a large number of features that are beyond the scope of a beginner-level academic project.

A simple Online or Offline status is useful for identifying the current condition of a device, but it does not provide a complete understanding of monitoring behaviour. Performance information, historical observations, risk interpretation, and comparison can provide additional context. Therefore, ChangeNet AI focuses on implementing a simplified version of these concepts. The project does not attempt to replace enterprise network monitoring platforms. Instead, it provides an academic environment for understanding network device management, monitoring visualization, alerts, and rule-based analysis.

1.3 PROBLEM STATEMENT

Monitoring network devices is important for understanding network availability and performance. However, beginners may find professional network monitoring systems difficult to configure and understand because of their extensive functionality. A basic device list showing only Online or Offline status also provides limited information. Users may need additional information such as monitoring measurements, traffic trends, risk levels, stability indicators, and simple interpretations of device behaviour.

The problem addressed by ChangeNet AI is therefore to develop a lightweight and user-friendly network monitoring dashboard that demonstrates device management, monitoring visualization, alert presentation, stability analysis, device comparison, and reporting without requiring complex infrastructure. The system should be simple enough for academic demonstration while still providing meaningful analytical information beyond a basic status indicator.

1.4 OBJECTIVES OF THE PROJECT

The main objectives of ChangeNet AI are:

1. To develop a simple web-based network monitoring dashboard.
2. To provide an administrative entry interface for accessing the application.
3. To provide predefined demonstration devices for easy project evaluation.
4. To allow users to add and manage manually registered network devices.
5. To display device information such as name, IP address, type, and status.
6. To provide simplified monitoring metrics such as CPU, memory, latency, packet loss, incoming traffic, and outgoing traffic.
7. To visualize monitoring information using a traffic graph and recent monitoring history.

8. To provide a centralized Alert Center for displaying sample network events.
9. To calculate a simple rule-based Device Stability Score.
10. To classify device conditions using a Change Risk Indicator.
11. To provide Smart Change Insight based on available device information.
12. To provide a Device Health Timeline for viewing recent status observations.
13. To provide a Compare Devices feature for comparing two devices.
14. To generate a Monitoring Report containing important device information.
15. To maintain a simple and responsive user interface suitable for academic demonstration.
16. To demonstrate networking concepts using lightweight web technologies without requiring a database

1.5 PROPOSED SYSTEM

The proposed ChangeNet AI system is a lightweight web-based network monitoring and device analysis platform. The application begins with a welcome interface followed by an administrative login section. After entering the login information, the user is provided with access to the main dashboard.

- **Dashboard** provides navigation to different functional sections, including Overview, Devices, Monitoring, Alerts, Smart Analysis, Compare Devices, and Monitoring Report.
- **Device Management** module provides manually maintained network device information. Users can add new devices by entering a device name, IP address, device type, and status. The system validates the IPv4 address and prevents duplicate IP addresses.
- **Network Monitoring** module provides simulated monitoring information including CPU utilization, memory utilization, latency, packet loss, incoming traffic, and outgoing traffic. A graph is used to display incoming and outgoing traffic trends. Recent measurements are presented in a monitoring history section.
- **Alert Center** provides sample network events such as router response delay, switch port warning, and device disconnection. The alerts are presented with priority information to demonstrate how important events can be highlighted.

- **Smart Analysis** module provides Change Risk Indicator, Device Health Timeline, and Smart Change Insight. These features use simple rule-based logic to provide understandable information about device conditions.
- **Compare Devices** module allows two devices to be selected and compared using status, stability score, and risk level.
- **Monitoring Report** module generates a concise summary containing device statistics, stability information, risk information, and report generation time.

The proposed system is intentionally lightweight and does not require a database or complex external framework.

1.6 BENEFITS OF THE PROJECT

ChangeNet AI provides the following benefits:

- Provides a simple and user-friendly network monitoring interface.
- Helps students understand basic network monitoring concepts.
- Provides a centralized dashboard for device information.
- Allows manual addition and management of network devices.
- Provides visual monitoring information through graphs and metrics.
- Demonstrates alert presentation using sample network events.
- Provides a simple Device Stability Score.
- Provides Change Risk classification.
- Provides understandable Smart Change Insight.
- Allows comparison between two devices.
- Provides a device health timeline.
- Generates a presentation-ready monitoring report.
- Does not require a database.
- Uses lightweight web technologies.
- Is easy to run and demonstrate locally.
- Provides a foundation for future real-time network monitoring improvements.

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

Network monitoring is an important area of computer networking that focuses on observing the availability, behaviour, and performance of network-connected devices. Monitoring systems help administrators identify unavailable resources, observe performance conditions, recognize network events, and understand changes in device behaviour. Modern monitoring platforms may use techniques such as ICMP checks, agents, SNMP, service monitoring, performance metrics, and historical data collection. Information obtained from monitoring systems is commonly presented through dashboards, charts, alerts, and reports. The study of network monitoring systems provides a foundation for designing ChangeNet AI. The project applies basic monitoring concepts in a simplified academic environment and combines monitoring visualization with rule-based device analysis.

2.2 RELATED WORK

Existing network monitoring systems commonly provide device availability monitoring, performance monitoring, event detection, alert generation, historical analysis, dashboards, and reporting.

- **Zabbix** is a widely used monitoring platform capable of monitoring network devices, servers, applications, and infrastructure components. It provides dashboards, metrics, alerting, and historical monitoring capabilities.
- **Nagios** is another established monitoring platform that provides host and service monitoring, notifications, and status information. It is widely used for infrastructure and network monitoring.
- Other monitoring solutions also provide graphical dashboards and performance information, allowing administrators to observe network conditions through charts and historical information.

These systems demonstrate that effective monitoring involves more than simply displaying an Online or Offline state. Monitoring information can be combined with historical data, alerts, performance measurements, and analysis to provide better understanding of system behaviour. ChangeNet AI adopts these general concepts at a much smaller scale.

Instead of attempting to provide enterprise-level monitoring, the project concentrates on manually maintained devices, simplified monitoring visualization, alert presentation, stability scoring, risk classification, device comparison, and report generation.

2.3 INFERENCE FROM RELATED WORK

The study of existing network monitoring approaches indicates that network monitoring can be divided into several important areas, including device management, status observation, performance monitoring, alerting, historical analysis, and reporting. The study also indicates that historical and performance information can provide additional context beyond a basic device status. Based on these observations, ChangeNet AI combines a simple device management interface with monitoring visualization and rule-based analysis. The project introduces a Device Stability Score to provide a simple numerical representation of device condition. A Change Risk Indicator further classifies the condition into understandable risk levels. Smart Change Insight provides a textual interpretation, while Device Health Timeline and Compare Devices provide additional ways to understand device information. The project is intentionally lightweight so that these concepts can be understood and demonstrated without requiring complex enterprise monitoring infrastructure.

CHAPTER 3

SYSTEM DESIGN

3.1 INTRODUCTION

The system design of ChangeNet AI defines the structure and interaction of the user interface, application logic, monitoring module, analysis components, and lightweight backend environment. The system follows a modular approach in which different sections are responsible for different functions.

The major components include:

- Welcome Interface
- Admin Login
- Dashboard
- Device Management
- Network Monitoring
- Alert Center
- Smart Analysis
- Device Comparison
- Monitoring Report

The application is implemented using HTML, CSS, and JavaScript. Node.js is included as a lightweight local backend environment.

3.2 SYSTEM ARCHITECTURE

The ChangeNet AI architecture consists of a presentation layer, application logic layer, monitoring layer, and analysis layer.

- **Presentation layer** provides the welcome page, login interface, dashboard, navigation, device interface, monitoring interface, alert center, analysis pages, comparison interface, and reporting interface.
- **Application logic layer** is implemented using JavaScript. It manages device information, user interactions, validation, dashboard statistics, monitoring simulation, graph rendering, analysis calculations, comparison, and report generation.

- **Monitoring layer** generates simplified demonstration values for CPU utilization, memory utilization, latency, packet loss, incoming traffic, and outgoing traffic. These values are simulated for demonstration purposes and are not collected directly from physical network devices in the current version.
- **Analysis layer** calculates a rule-based stability score and derives change-risk information and Smart Change Insight.

The overall architecture can therefore be described as:

User → Web Interface → JavaScript Application Logic → Device and Monitoring Data → Analysis → Dashboard / Alerts / Insights / Reports

Node.js is included to provide a lightweight local backend environment and a foundation for future integration of actual network monitoring services.

3.3 SYSTEM REQUIREMENTS

The system requires a basic computer capable of running a modern web browser and Node.js.

Hardware Requirements

The application can be developed and demonstrated using:

- Dual-core processor or above
- Minimum 4 GB RAM
- At least 500 MB available storage
- Display resolution of approximately 1366×768 or above
- Local network connectivity when required for future real monitoring functionality

Software Requirements

- **Operating System:** Windows, Linux, or macOS
- **HTML:** Used for structuring the web interface.
- **CSS:** Used for styling, layout, responsive design, and visual presentation.
- **JavaScript:** Used for application logic, device management, monitoring simulation, graph generation, analysis, comparison, and report generation.
- **Java:** Used for rule-based Analysis
- **Node.js:** Used as a lightweight local backend runtime.

3.4 DATA FLOW DIAGRAM

The user first interacts with the ChangeNet AI welcome interface and proceeds to the administrative login section. After successful entry of login information, the dashboard becomes available.

- The user can then access the **Device Management module** to view predefined devices or add new devices. When a device is added, the application validates the provided device information, including the IPv4 address and duplicate IP condition.
- The **Monitoring module** generates simplified monitoring values such as CPU utilization, memory utilization, latency, packet loss, incoming traffic, and outgoing traffic. These values are updated periodically during the monitoring demonstration. The monitoring information is displayed using metric cards, a traffic graph, and a recent monitoring history section.
- The **Alert Center** displays predefined sample network events and their corresponding priority levels.
- The available device information is then processed by the Smart Analysis module. The system calculates a rule-based stability score and derives a change-risk classification. The resulting information is displayed through Smart Change Insight.
- The **Compare Devices** module receives information from the device collection and displays a comparison between two selected devices.

Finally, the Monitoring Report module summarizes the available device and stability information in a presentation-ready format.

The simplified data flow is:

User Input → Login → Device Management → Monitoring Data → Visualization → Alert Display → Stability Analysis → Risk Analysis → Smart Insight → Device Comparison → Monitoring Report

CHAPTER 4

PROJECT DESCRIPTION

4.1 METHODOLOGIES

The project follows an incremental development methodology.

The system was developed by first establishing the basic user interface and navigation structure. Device management functionality was then introduced, followed by monitoring visualization, alert presentation, analysis features, comparison, and reporting.

Frontend Development

HTML is used to define the structure of the application. CSS is used to provide the visual design, layout, responsive behaviour, cards, tables, navigation elements, status indicators, and dashboard presentation. JavaScript is used to implement the application logic and user interaction.

Device Management Methodology

The system maintains a collection of predefined demonstration devices and allows additional devices to be added during use. When a new device is entered, the system validates the device name, IPv4 address, and device type. Duplicate IP addresses are rejected.

Monitoring Methodology

The current implementation uses simulated monitoring data to demonstrate network monitoring concepts.

The system periodically generates values for:

- CPU utilization
- Memory utilization
- Latency
- Packet loss
- Incoming traffic
- Outgoing traffic

These values are displayed through monitoring cards and a graphical traffic trend. This approach allows the project to demonstrate monitoring concepts without requiring physical network devices or specialized network configuration.

Alert Methodology

The Alert Center currently uses predefined sample network events. These events represent conditions such as router response delay, switch port warning, and device disconnection. The events are presented with different priority levels to demonstrate how monitoring alerts can be categorized and displayed.

Stability Analysis Methodology

The Device Stability Score uses a simple rule-based calculation. It produces a numerical value within the range of 0 - 100.

The score is intended as a demonstration indicator rather than a scientifically validated network reliability measurement.

Smart Analysis Methodology

The Smart Change Insight uses rule-based logic to convert the available device status and calculated stability information into a simple textual interpretation.

The current implementation does not use a machine-learning model.

Reporting Methodology

The Monitoring Report module collects the current device information and calculates summary values such as total devices, online devices, offline devices, most stable device, device requiring attention, and change-risk level.

4.2 MODULE DESCRIPTION

1. Admin Login

- The Admin Login module provides the entry interface for the application.
- The user enters an Admin ID and password before accessing the dashboard. The current implementation is intended for demonstration purposes and accepts any non-empty ID and password.
- Therefore, the login should be considered a demo authentication interface rather than a production-grade security system.

2. Dashboard

It presents important information such as:

- Total number of devices
- Number of online devices
- Number of offline devices
- Alert information
- Recent monitoring information

The dashboard allows users to quickly understand the current demonstration environment.

3. Device Management

Each device contains information such as:

- Device Name
- IP Address
- Device Type
- Current Status

Users can:

- View devices
- Add a new device
- Search devices
- Toggle device status
- Remove devices

The system also validates IPv4 addresses and prevents duplicate IP addresses.

4. Network Monitoring

The Network Monitoring module provides simplified network performance information.

The module displays:

- CPU utilization
- Memory utilization
- Latency
- Packet loss
- Incoming traffic
- Outgoing traffic

The system generates simulated values at regular intervals. The values are visualized using a network usage graph. The Recent Monitoring History section displays recent measurements including time, CPU, memory, latency, incoming traffic, and outgoing traffic. The current monitoring module is intended for academic demonstration and does not directly collect real-time telemetry from physical network devices.

5. Alert Center

The Alert Center provides a centralized location for displaying network-related events.

The current version contains demonstration events such as:

- Router response delay
- Switch port warning
- Device disconnected

Each event contains priority information to demonstrate how different levels of network conditions can be visually represented. The current alerts are sample events rather than dynamically generated alerts from physical network monitoring.

6. Smart Analysis

The Smart Analysis module combines several simple analytical features.

It provides:

- Change Risk Indicator
- Device Health Timeline
- Smart Change Insight

These features are designed to make basic monitoring information easier to interpret.

7. Change Risk Indicator

The Change Risk Indicator classifies the available device condition into:

- Low Risk
- Medium Risk
- High Risk

The classification is derived from the rule-based stability information available in the application.

8. Device Health Timeline

- The Device Health Timeline presents recent status observations in chronological form.
- It provides a simple visual representation of device status conditions such as Online and Offline.
- The feature is intended to help users understand how a device condition can be presented over a sequence of observations.

9. Device Stability Score

- The Device Stability Score provides a numerical indication of device stability. The score ranges from 0 to 100A higher value represents a more stable condition, while a lower value represents a condition requiring greater attention.
- The score is calculated using simple rule-based logic and is intended for demonstration rather than as a formal network reliability metric.

10. Smart Change Insight

- Smart Change Insight converts the available device status and stability information into a simple textual explanation.
- For example, a device with a relatively high stability score may be described as reasonably stable, while a device with a lower score may be identified as requiring attention.
- The current implementation uses rule-based logic rather than machine learning.

11. Compare Devices

The Compare Devices module allows users to select two devices and compare their information.

The comparison includes:

- Current Status
- Stability Score
- Risk Level

The system also indicates which device currently has the higher stability score.

12. Monitoring Report

The Monitoring Report module generates a presentation-ready summary of the monitored environment.

The report includes:

- Total Devices
- Online Devices
- Offline Devices
- Most Stable Device
- Stability Score
- Device Requiring Attention
- Change Risk
- Report Generation Time

4.3 IMPLEMENTATION AND WORKING

ChangeNet AI is implemented using HTML, CSS, and JavaScript with Node.js included as a lightweight backend environment.

- Application begins with a welcome interface and provides an administrative login section. After entering non-empty login information, the user is directed to the main dashboard.
- The dashboard provides access to device management, monitoring, alerts, smart analysis, device comparison, and reporting.
- The Device Management module initializes with predefined demonstration devices. Users can add new devices by entering a device name, valid IPv4 address, device type, and status.
- The application validates the IPv4 address and checks for duplicate IP addresses before adding a device.
- The monitoring system generates simulated values for CPU utilization, memory utilization, latency, packet loss, incoming traffic, and outgoing traffic. These values are periodically updated and displayed in the Network Monitoring section.
- The incoming and outgoing traffic values are visualized through a graph. The Recent Monitoring History section displays the latest measurements.
- The Alert Center presents predefined sample network events and priority information.
- The Smart Analysis section uses rule-based logic to calculate a stability score and corresponding risk level. It also provides Smart Change Insight and a Device Health Timeline.
- The Compare Devices module allows users to select two devices and compare their current status, stability score, and risk.
- The Monitoring Report module generates a summary of the current device environment and analysis results.

The implementation is intentionally lightweight and does not require a database or machine-learning framework.

CHAPTER 5

RESULTS AND DISCUSSION

The developed **ChangeNet AI** application provides a simple and user-friendly environment for demonstrating network monitoring and device analysis concepts. The system integrates device management, network monitoring, alert presentation, smart analysis, device comparison, and monitoring report generation within a single web-based platform. The Device Management module allows users to view predefined network devices and add new devices by providing information such as device name, IP address, device type, and status. The Network Monitoring module presents important monitoring parameters including CPU utilization, memory utilization, latency, packet loss, incoming traffic, and outgoing traffic in an organized dashboard. In the current implementation, these monitoring values are simulated for academic demonstration, allowing users to understand how network performance information can be collected and visualized without requiring physical network monitoring infrastructure. The monitoring graph provides a visual representation of traffic behaviour, while the Recent Monitoring History section presents the latest measurements in an organized format. The Alert Center demonstrates the presentation of network events such as router response delay, switch port warning, and device disconnection using different priority levels. The Smart Analysis module extends the basic monitoring functionality by providing Device Stability Score, Change Risk Indicator, Device Health Timeline, and Smart Change Insight. The Device Stability Score provides a simple rule-based indication of the observed device condition, while the Change Risk Indicator classifies the condition into different levels of attention. The Compare Devices module allows users to compare two devices based on their status, stability score, and risk level, while the Monitoring Report module provides a concise summary of important device and analysis information. Overall, the results demonstrate that ChangeNet AI can combine basic network monitoring concepts, visualization, and rule-based analysis in a lightweight and visually organized application suitable for academic demonstration and project evaluation.

CHAPTER 6

CONCLUSION AND FUTURE WORK

ChangeNet AI was successfully developed as a lightweight Network Monitoring and Device Analysis System that demonstrates important concepts of network device management, monitoring, visualization, and basic analytical processing. The system provides a centralized and easy-to-use interface through which users can access device information, monitoring metrics, alerts, smart analysis, device comparison, and monitoring reports. The application demonstrates network monitoring using parameters such as CPU utilization, memory utilization, latency, packet loss, incoming traffic, and outgoing traffic, with simulated values used in the current version to provide a practical demonstration environment. The project also extends basic monitoring functionality through the Device Stability Score, which provides a simple rule-based indication of device condition, and the Change Risk Indicator, which helps identify devices that may require greater attention. The Smart Change Insight feature further improves the understanding of monitoring information by converting available device conditions into a simple textual interpretation. In addition, the Device Health Timeline provides a chronological view of device conditions, while the Compare Devices feature allows users to evaluate two devices based on their status, stability, and risk information. The Monitoring Report provides a summarized representation of the available device and analysis information, making the application useful for demonstrations and presentations. The project is implemented using lightweight technologies such as HTML, CSS, JavaScript, and Node.js, making it relatively simple to understand, operate, and maintain. Although the present version uses simulated monitoring values and sample alert events, the developed system provides a strong foundation for future integration of real-time network monitoring, database storage, secure authentication, SNMP or ICMP-based monitoring, advanced analytics, and machine-learning-based anomaly detection. Overall, ChangeNet AI successfully demonstrates how web technologies and basic rule-based analysis can be combined to create a practical, accessible, and visually organized network monitoring platform for academic purposes.

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